The consequences of this thought tumbled forth quickly, and were immortalized in Einstein's review article. If you fall freely (for example, by jumping off a cliff), not only will you not feel your own weight, it will seem to you, in all respects, as though gravity had completely disappeared from your vicinity. For example, if you drop some rocks from your hand as you fall, you and the rocks will then fall together, side by side. If you look at the rocks and ignore your other surroundings, you cannot discern whether you and the rocks are falling together toward the ground below or are floating freely in space, far from all gravitating bodies. In fact, Einstein realized, in your immediate vicinity, gravity is so irrelevant, so impossible to detect, that *all* the laws of physics, in a small reference frame (laboratory) that you carry with you as you fall, must be the same as if you were moving freely through a universe without gravity. In other words, your small, freely falling reference frame is "equivalent to" an inertial reference frame in a gravity-free universe, and the laws of physics that you experience are the same as those in a gravity-free inertial frame; they are the laws of special relativity. (We shall learn later why the reference frame must be kept small, and that "small" means very small compared to the size of the Earth—or, more generally, very small compared to the distance over which the strength and direction of gravity change.)

As an example of the equivalence between a gravity-free inertial frame and your small, freely falling frame, consider the special relativistic law that describes the motion of a freely moving object (let it be a cannonball) in a universe without gravity. As measured in any inertial frame in that idealized universe, the ball must move along a straight line and with uniform velocity. Compare this with the ball's motion in our real, gravity-endowed Universe: If the ball is fired from a cannon on a grassy meadow on Earth and is watched by a dog who sits on the grass, the ball arcs up and over and falls back to Earth (Figure 2.2). It moves along a parabola (solid black curve) as measured in the dog's reference frame. Einstein asks that you view this same cannonball from a small, freely falling reference frame. This is easiest if the meadow is at the edge of a cliff. Then you can jump off the cliff just as the cannon is fired, and watch the ball as you fall.

As an aid in depicting what you see as you fall, imagine that you hold in front of yourself a window with twelve panes of glass, and you watch the ball through your window (middle segment of Figure 2.2). As you fall, you see the clockwise sequence of scenes shown in Figure 2.2. In looking at this